

IIT JAM Mathematics (MA)

Comprehensive Project Statistics & Analytics Report

22-Year Historical Archive (2005–2026) • 885 Official Examination Questions • Ecosystem Metrics

885

Official Questions
100% Verified Keys

22

Exam Years
2005 – 2026 (22 Yrs)

58

CBT Mock Tests
22 Years + 27 Eras + 9 Arch

59

LaTeX PDFs
Publication Quality

1. Executive Summary & Repository Overview

The **IIT JAM Mathematics (MA) CBT Platform and Examination Archive** is an open-access academic repository hosting all **885 official examination questions** spanning **22 consecutive years (2005–2026)**. Every question is digitized with its original high-resolution cropped screenshot, tagged by domain, classified by subtopic, and paired with verified master answer keys (including numerical tolerance ranges).

Core Pillars of the Platform:

- **Authentic CBT Simulator:** 58 interactive mock tests featuring live timer, marking scheme, palette, calculator, review flags, and analytics.
- **Custom Test & Instant PDF Generator:** Client-side LaTeX PDF compilation engine for tests with 1 to 100 questions.
- **Formula Cheat Sheet Library:** 14 dedicated high-yield formula sheets.
- **Master Historical Archive:** Complete 35.2 MB original examination papers archive.

Project Architecture at a Glance

Questions Digitized:	885
Examination Years:	22 (2005–2026)
Subject Domains:	3 Core + Legacy
Syllabus Subtopics:	9 Categories
Screenshot Cards:	885 PNG Cards
Formula Cheat Sheets:	14 Topic PDFs
Pre-compiled PDFs:	59 Formatted PDFs
Client PDF Engine:	Instantaneous

2. Examination Eras & Pattern Evolution (2005–2026)

Over the 22-year span from 2005 to 2026, the IIT JAM Mathematics examination transitioned through three major testing paradigms:

Testing Era	Year Span	Papers	Questions	MCQ	MSQ	NAT	Examination Structure
Recent CBT Era	2022–2026	5 Yrs	300 Qs	150	50	100	60 Qs · 100 Marks (Sec A: 30, B: 10, C: 20)
Established CBT Era	2015–2021	7 Yrs	420 Qs	210	70	140	60 Qs · 100 Marks (Sec A: 30, B: 10, C: 20)
Classic Paper Era	2005–2014	10 Yrs	165 Qs	165	0	0	Objective & Subjective Format
Total / Summary	22 Years	22	885 Qs	525	120	240	100% Fully Verified Keys

Key Observations on Pattern Evolution:

- **The CBT Revolution (2015–Present):** Standardized 60-question, 100-mark format comprising Section A (30 MCQs), Section B (10 MSQs), and Section C (20 NATs).
- **Negative Marking Scheme:** Section A carries $-\frac{1}{3}$ penalty for 1-mark MCQs and $-\frac{2}{3}$ penalty for 2-mark MCQs. Sections B (MSQ) and C (NAT) carry strictly zero negative marking.
- **Classic Era (2005–2014):** Featured diverse formats including 15-question high-value (6 marks each) objective problem sets and transitional 35-question tests (2014).

3. Subject Domain & Subtopic Weightage Analysis

All 885 questions have been classified into **3 primary syllabus subjects**, **9 official subtopics**, and a dedicated **Legacy / Out of Syllabus** category for topics no longer in the current syllabus (e.g., Mechanics and Vector Calculus).

Subtopic Name	Subject Domain	Qs	Marks	MCQ	MSQ	NAT	% of Total
1.1 Sequences & Series of Real Numbers	Real Analysis	131	292	85	15	31	14.8%
1.2 Functions of One Real Variable	Real Analysis	155	329	85	27	43	17.5%
Subtotal: Real Analysis	Domain 1	286	621	170	42	74	32.3%
2.1 Functions of Several Variables	Multivariable & ODE	78	152	44	13	21	8.8%
2.2 Integral Calculus	Multivariable & ODE	66	136	35	3	28	7.5%
2.3 Differential Equations (ODE)	Multivariable & ODE	113	255	70	12	31	12.8%
Subtotal: Multivariable & ODE	Domain 2	257	543	149	28	80	29.0%
3.1 Basic Algebra	Linear Algebra & Algebra	3	6	0	1	2	0.3%
3.2 Matrices & Systems of Linear Eq.	Linear Algebra & Algebra	75	155	43	9	23	8.5%
3.3 Finite Dimensional Vector Spaces	Linear Algebra & Algebra	76	184	44	12	20	8.6%
3.4 Groups & Subgroups	Linear Algebra & Algebra	101	214	56	16	29	11.4%
Subtotal: Linear Algebra & Algebra	Domain 3	255	559	143	38	74	28.8%
Legacy / Out of Syllabus (2005–2021)	Legacy Topics	87	277	63	12	12	9.8%
Grand Total (All Questions)	Master Dataset	885	2,000	525	120	240	100.0%

4. Question Type & Mark Distribution Breakdown

MCQ (Multiple Choice)

- **Total Count:** 525 Qs (59.3%)
- **1-Mark MCQs:** 130 Questions
- **2-Mark MCQs:** 275 Questions
- **Classic 6-Mark MCQs:** 120 Qs
- **Marking Penalty:** $-\frac{1}{3}$ & $-\frac{2}{3}$

MSQ (Multiple Select)

- **Total Count:** 120 Qs (13.6%)
- **Marks per Question:** 2 Marks
- **Total Marks in MSQs:** 240 Marks
- **Full Credit:** All correct options
- **Negative Marking:** None (0 marks)

NAT (Numerical Answer)

- **Total Count:** 240 Qs (27.1%)
- **1-Mark NATs:** 120 Questions
- **2-Mark NATs:** 120 Questions
- **Range Tolerances:** Official bounds
- **Negative Marking:** None (0 marks)

Subject Domain Observations:

- **Real Analysis** is the single largest subject domain with **32.3%** of all questions and **621 marks**, led by **Functions of One Real Variable (155 Qs)** and **Sequences & Series (131 Qs)**.
- **Linear Algebra and Multivariable Calculus** carry nearly equal weightage (**28.8%** and **29.0%** respectively), ensuring balanced cross-domain preparation.

5. Annual Examination Matrix (2005–2026)

Year	Era	Qs	Marks	MCQ	MSQ	NAT	Examination Pattern
2005	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2006	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2007	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2008	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2009	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2010	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2011	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2012	Classic Paper	15	90	15	0	0	15 Objective Problems (6M each)
2013	Classic Paper	10	20	10	0	0	10 Objective Questions (2M each)
2014	Classic Paper	35	60	35	0	0	35 Objective Questions (1M / 2M)
2015	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2016	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2017	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2018	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2019	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2020	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2021	CBT Era	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2022	Recent CBT	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2023	Recent CBT	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2024	Recent CBT	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2025	Recent CBT	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
2026	Recent CBT	60	100	30	10	20	CBT (30 MCQ + 10 MSQ + 20 NAT)
Total	22 Yrs	885	2000	525	120	240	Complete 22-Year Archive

6. CBT Simulator Ecosystem & Asset Inventory

58 Interactive CBT Mock Tests

- 22 Year-Wise Tests (2005–2026): Full 180-min official exams.
- 27 Era Subtopic Tests: 3 Eras × 9 Topics.
- 9 Archive Subtopic Tests: 22-Year all-time topic mastery.
- Instant Custom Generator: 1–100 question custom tests.

Study Materials & Asset Library

- 14 Formula Cheat Sheets: High-yield domain sheets.
- 59 Pre-compiled LaTeX PDFs: Ready-to-print papers.
- 885 Cropped Screenshots: High-res PNG problem cards.
- Official Master Archive: 35.2 MB 22-year paper PDF.

Report Generation Metadata: Compiled automatically from the verified master dataset. All data, answer keys, and assets are hosted on the open-source repository at vaibhavgeometer.github.io.